

Figure 9: FSInt non-stripped binaries

- Then, files of the same size are checked to ensure that these are not hard links.
- Subsequently, it calculates the checksum of the file using `md5sum`.
- Finally, to guard against *md5sum collisions*, FSInt rechecks signatures of any remaining files using the *sha1sum* checksum.

Installed packages

This tool lists the installed packages according to their sizes. It supports the Debian package manager (dpkg), the Red Hat package manager (RPM) and the Packman package manager. When a package is selected, a description of it is shown below the *results* window in a gray box. Figure 3 shows the results of the operation.

Bad names

Another common form of file system lint is bad names. Even though they do not eat up disk space, they may be difficult to use or move. The *Bad names* tool searches all the files and inspects their naming conventions. You can set the sensitivity level by using the slider bar at the top of the selection window. Level 1 implies the least strict while Level 4 implies a strict POSIX check. Besides the slider, there is a check box, which allows you to select UTF-8 checking. The image in Figure 4 shows the results of the operation.

Name clashes

Another form of file system lint involves files that have identical or similar names. This usually does not cause any significant problems for the user other than slight inconvenience. But finding files with name clashes can be of great help when dealing with multiple versions of files (Figure 5).

Temp files

Temporary files eat up a lot of hard disk space. These are created while editing files, while running some programs,

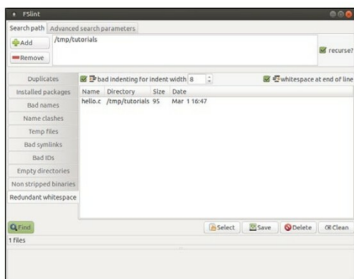


Figure 10: FSInt redundant white spaces

or when a program wishes to report and save a problem. Removing these temporary files can free up valuable space, and FSInt can find these files so that they can be purged.

It is also possible to direct the program to find only files of a specified minimum 'age'—like those modified in the last 'N' number of days, for instance. When the minimum age is set to 0, FSInt reports all temporary files. FSInt provides a check box *core file mode?* which enables a more thorough search for the core files. Figure 6 shows the results of the operation.

Bad symlinks

Symbolic links are widely used in GNU/Linux. But broken symbolic links cause great frustration for the user. FSInt looks for symbolic links that have some kind of problem and immediately reports them. It specifically looks for the following problems.

- Dangling:** In this type, a broken symbolic link points to a file that no longer exists in the file system.
- Suspect:** In this type, symbolic links point to a file below their directory structure.
- Relative:** In this type, a symbolic link points to a path that is determined by the current location.
- Absolute:** In this type, a symbolic link points to a path that is determined by the full path.



Note: It seems that there is a bug in the GUI tool for this operation. Because, though the command line tool (`/usr/share/fsint/fsint/findbl`) reports broken links correctly, the GUI is unable to show correct results.

Bad IDs

GNU/Linux assigns a positive integer number as 'user id' to each user. When the user moves files between multiple computers, a file will occasionally end up with a user ID that the current system cannot resolve to that user. The most common forms of bad IDs are generated when we extract